

Coefficient mapping for MadDM

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ABSTRACT: In this note I discuss the python script called “alpha_map.py”. It is a functionality that takes the effective coefficients that MadDM can already determine, and maps them onto the non-relativistic operators for direct detection.

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1 Background

In the MadDM code [1] there is an algorithm that determines the coefficients to the effective interactions that are relevant for direct detection experiments. Expressed in terms of a Lagrangian, this is

$$\mathcal{L}_{\text{eff}} = \sum_{q,s} \hat{\mathcal{O}}_{q,s}(x) + \hat{\mathcal{O}}'_{q,s}(x)$$

where q and s are labels tell you what quarks are being considered and whether the operator is even (e) or odd (o) under a $q \rightarrow \bar{q}$ transformation.

2 Fermion dark matter

In MadDM, one defines the dark matter particle that their considering so the spin of the particle is set. From that, we can list the operators for fermion dark matter as outlined in Ref. [1]¹

	Even	Odd
SI	$\bar{\psi}_x \psi_x \bar{\psi}_q \psi_q$	$(\bar{\psi}_x \gamma^\mu \psi_x) (\bar{\psi}_q \gamma_\mu \psi_q)$
SD	$(\bar{\psi}_x \gamma^\mu \gamma^5 \psi_x) (\bar{\psi}_q \gamma_\mu \gamma^5 \psi_q)$	$-\frac{1}{2} (\bar{\psi}_x \sigma^{\mu\nu} \psi_x) (\bar{\psi}_q \sigma_{\mu\nu} \psi_q)$

These operators are can be related to the operators in Ref. [2] which is meant to be a direct detection relevant Standard Model Effective field theory (SMEFT) basis,

$$\begin{aligned}
\mathcal{O}_e^{\text{SI}} &= \frac{1}{m_f} Q_{s,f}^{(7)}, \quad Q_{s,f}^{(7)} = (\bar{\chi} \chi) (\bar{f} f), \\
\mathcal{O}_o^{\text{SI}} &= Q_{1,f}^{(5)} = (\bar{\chi} \gamma^\mu \chi) (\bar{f} \gamma_\mu f), \\
\mathcal{O}_e^{\text{SD}} &= Q_{4,f}^{(5)} = (\bar{\chi} \gamma^\mu \gamma^5 \chi) (\bar{f} \gamma_\mu \gamma^5 f), \\
\mathcal{O}_o^{\text{SD}} &= \frac{1}{m_f} Q_{q,f}^{(7)}, \quad Q_{q,f}^{(7)} = m_f (\bar{\chi} \sigma^{\mu\nu} \chi) (\bar{f} \sigma_{\mu\nu} f).
\end{aligned}$$

where I've separated the spin-independent (SI) and spin-dependent (SD) operators because the MadDM algorithm is able to do that.

References

- [1] M. Backović, A. Martini, O. Mattelaer, K. Kong, and G. Mohlabeng, *Direct Detection of Dark Matter with MadDM v.2.0*, *Phys. Dark Univ.* **9-10** (5, 2015) 37–50, [[arXiv:1505.04190](#)].
- [2] F. Bishara, J. Brod, B. Grinstein, and J. Zupan, *DirectDM: a tool for dark matter direct detection*, [arXiv:1708.02678](#).

¹there are more, but I think they are in principle subdominant. Perhaps we should include them later on.